ZJU-UIUC Institute

ECE 313– Final Examination
Probability with Engineering Applications

Instructor: Prof. Piao Chen

9:00-12:00, 26 May 2024

Name: ______________________________
ID Number: ______________________________

Instructions

- You may not use any books or notes other than three two-sided sheets of handwritten A4 paper.
- You may not use any electronic devices other than a non-programmable calculator.
- When you are asked to "calculate," "determine," or "find," this means providing closed-form expressions (i.e., without summation or integration signs).
- Show all your work to receive full credit for your answers.
- Neatness counts. If we are unable to read your work, we cannot grade it.
- Turn in your entire booklet once you are finished. Scratch papers and normal table are provided in the end.
- This exam contains 14 pages and 9 questions. Total of points is 100.

Distribution of Marks

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	30	6	10	6	18	6	6	6	12	100
Score:										

1. (30 points) Select either True or False for each statement below and justify your response (an answer without justification will receive no credit). In order to discourage guessing, 1 point will be deducted for each incorrect answer and/or justification (no penalty or gain for blank answers). A net negative score will reduce your total exam score.

   (a) Consider three events A, B and C such that $P(ABC) = P(A)P(BC) > 0$ and $P(AB) = P(A)P(B)$.

   TRUE FALSE

   ☐ ☐ $P(ABC) = P(A)P(B)P(C)$.

   ☐ ☐ $P(C|AB) = P(C|B)$.

   ☐ ☐ If $P(B|C) < P(C|B)$, then $P(B) < P(C)$.

   (b) Suppose that $P(A) = 0.4, P(B) = 0.3$ and $P((A \cup B)^c) = 0.42$.

   TRUE FALSE

   ☐ ☐ $P(AB) > P(A)P(B)$

   ☐ ☐ $P((AB)^c) = 0.84$.

(c) Suppose that T is a nonnegative random variable with failure rate function $h(t)$.

TRUE FALSE

☐ ☐ If h is a constant then T follows a normal distribution.

☐ ☐ If the failure rate function of another random variable S is $2h(t)$, then it must be that $E(T) = 2E(S)$.

(d) Consider a Poisson process of rate 1. Let T_1 be the time of the first count and T_2 be the time of second count.

TRUE FALSE

☐ ☐ The number of arrivals between T_1 and T_2 is a Poisson random variable.

☐ ☐ $T_2 - T_1$ has the exponential distribution with rate 2.

(e) Suppose (X, Y) are jointly Gaussian with $\mu_X = 1, \mu_Y = 0, \sigma_X^2 = 1, \sigma_Y^2 = 4$ and $\rho_{XY} = \frac{1}{8}$.

TRUE FALSE

☐ ☐ Let $W = X + \alpha Y + \beta$. When $\alpha = -4$, no matter what value β takes, X and W are independent.

2. Tom is diagnosed with an uncommon disease. He knows that there is only a 2% chance of getting it. Use the letter D for the event "Tom has the disease" and T for "the test says so". It is known that the test is imperfect: $P(T|D) = 0.95$ and $P(T^c|D^c) = 0.98$.

   (a) (3 points) Given that Tom tests positive, what is the probability that he really has the disease?

   (b) (3 points) Tom obtains a second opinion: an independent repetition of the test. He tests positive again. Given this, what is the probability that he really has the disease?

3. Let X_1, X_2 and X_3 be three independent $Geo(p)$ distributed random variables, and let $Z = X_1 + X_2 + X_3$.

   (a) (3 points) Derive the probability mass function p_Z of Z.

   (b) (3 points) Use (a) to show that

   $$p^2 \left[E(X_1^2) + E(X_1)\right] = 2.$$

   (c) (4 points) Use (b) to derive $E(X_1^2)$ and $Var(X_1)$.

4. Let η be an unknown real number, and let the joint probabilities $P(X = a, Y = b)$ of the discrete random variables X and Y be given by the following table:

	a		
b	-1	0	1
4	$\eta - \frac{1}{8}$	$\frac{1}{8} - \eta$	0
5	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{8}$
6	$\eta + \frac{1}{8}$	$\frac{1}{16}$	$\frac{3}{8} - \eta$

(a) (3 points) Which are the values η can attain?

(b) (3 points) Is there a value of η for which X and Y are independent?

5. Let the pair (X, Y) be drawn arbitrarily from the triangle Δ with vertives (0,0),(0,1), and (1,1).
   (a) (2 points) Show that the joint distribution function F of the pair (X, Y) satisfies

$$F(a,b) = \begin{cases} 0 & \text{for } a \text{ or } b \text{ less than } 0 \\ a(2b-a) & \text{for } (a,b) \text{ in the triangle } \Delta \\ b^2 & \text{for } b \text{ between 0 and 1 and } a \text{ larger than } b \\ 2a - a^2 & \text{for } a \text{ between 0 and 1 and } b \text{ larger than 1} \\ 1 & \text{for } a \text{ and } b \text{ larger than 1.} \end{cases}$$

   (b) (2 points) Determine the joint probability density function f of the pair (X, Y).
   (c) (2 points) Derive the marginal densities of X and Y.
   (d) (4 points) An arbitrary point (U, V) is drawn from the unit square $[0,1] \times [0,1]$. Show that $\min\{U, V\}$ has the same distribution as X and that $\max\{U, V\}$ has the same distribution as Y.
   (e) (8 points) Derive $E(X)$, $Var(X)$, $E(Y)$, $Var(Y)$, $Var(X+Y)$ and $Cov(X, Y)$.

This page is intentionally left blank to accommodate answers for Question 5.

6. A factory produces links for heavy metal chains. The research lab of the factory models the length (in cm) of a link by the random variable X, with expected value $E(X) = 6$ and variance $Var(X) = 0.04$. The length of a link is defined in such a way that the length of a chain is equal to the sum of the lengths of its links. The factory sells chains of 60 meters; to be on the safe side 1004 links are used for such chains. The factory guarantees that the chain is not shorter than 60 meters. If by chance a chain is too short, the customer is reimbursed, and a new chain is given for free.

   (a) (3 points) Give an estimate of the probability that for a chain of at least 60 meters more than 1004 links are needed. For what percentage of the chains does the factory have to reimburse clients and provide free chains?

   (b) (3 points) The sales department of the factory notices that it has to hand out a lot of free chains and asks the research lab what is wrong. After further investigations the research lab reports to the sales department that the expectation value 6 is incorrect, and that the correct value is 5.98 (cm). Do you think that it was necessary to report such a minor change of this value?

7. Suppose we have a random sample $X_1, \cdots, X_n$ from an $Exp(\lambda)$ distribution. We want to estimate the expectation $1/\lambda$. We know that

$$\bar{X}_n = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$$

is an unbiased estimator of $1/\lambda$. Let us consider more generally estimators T of the form

$$T = c \cdot (X_1 + X_2 + \cdots + X_n),$$

where c is a real number. We are interested in the MSE of these estimators and would like to know whether there are choices for c that yield a smaller MSE than the choice $c = 1/n$.

(a) (3 points) Compute MSE(T) for each c.

(b) (3 points) For which c does the estimator perform best in the MSE sense? Compare this to the unbiased estimator $\bar{X}_n$ that one obtains for $c = 1/n$.

8. You are studying the population size of cities in hundreds of thousands. For a place to be called a city, there should be some lower limit $\delta + 1$ to the population size and you assume that the population size will follow a Pareto distribution after that limit. So you model that a random sample $Y_1, \cdots, Y_n$ of cities will follow a distribution given by $Y = \delta + X$ where $X \sim Par(\alpha)$. In other words, the Y_i's are independent and each have probability density function given by

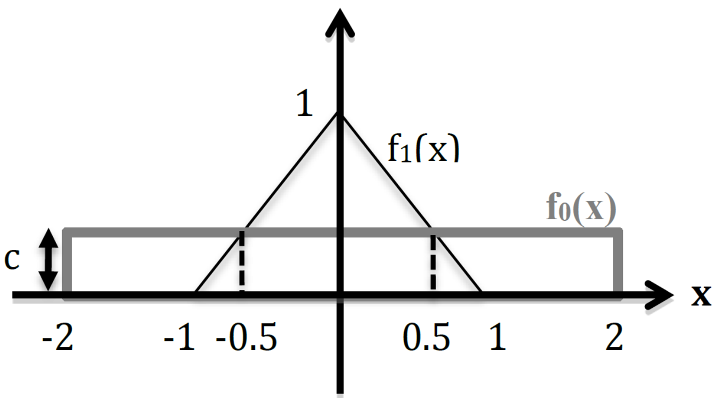
$$f(x) = \begin{cases} 0 & \text{if } x < 1 + \delta \\ \frac{\alpha}{(x-\delta)^{\alpha+1}} & \text{if } x \geq 1 + \delta. \end{cases}$$

Suppose you obtained a dataset $y_1, \cdots, y_n$ from this random sample.

(a) (3 points) Find the maximum likelihood estimators for the pair (α, δ).

(b) (3 points) Suppose $n = 5$ and $\alpha = 3$ then what is the mean squared error for the maximum likelihood estimator for δ expressed as a function of δ?

9. As shown in the following figure, an observation X is drawn from a triangular distribution $f_1(x)$ with range $[-1, 1]$ if hypothesis H_1 is true, and is drawn from a uniform distribution $f_0(x)$ with range $[-2, 2]$ if hypothesis H_0 is true. Assume that the prior probability $P(H_0)$ is 0.6.



(a) (4 points) Describe the maximum likelihood (ML) decision rule in terms of observation X.

(b) (3 points) Calculate the miss detection probability, false alarm probability and error probability for the ML rule.

(c) (5 points) Repeat (a) and (b) for the MAP decision rule.

This page is intentionally left blank to accommodate answers for Question 9.

Standard Normal Distribution Table (Right-Tail Probabilities)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002